EE-B-575D
April 7, 1980
SUPERSEDING
Fed. Spec. EE-B-575C
December 7, 1976

FEDERAL SPECIFICATION

BOUILLON (SOUP AND GRAVY BASES)

This specification was approved by the Secretary, United States Department of Agriculture, for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cubed and granulated vegetable protein bouillon, and powder, cube and paste, beef, chicken and ham flavored, soup and gravy bases, for use by all Federal agencies.

1.2 Classification.

1.2.1 Types, classes and styles. The bouillon (soup and gravy bases) covered by this specification shall be of the following types, classes and styles, as specified (see 6.1):

Type I	-	Beef flavor
Class 1	-	Regular
Style A	-	Powder
Style B	-	Cube
Style C	-	Paste
Class 2	-	Low sodium
Style A	-	Powder
Style B	-	Cube
Style C	-	Paste
Type II	-	Chicken flavor
Class 1	-	Regular
Style A	-	Powder
Style B	-	Cube
Style C	-	Paste

FSC 8935

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Class 2 - Low Sodium

Style A - Powder
Style B - Cube
Style C - Paste

Type III - Ham flavor

Class 1 - Regular

Style A - Powder

Type IV - Vegetable protein bouillon

Class 1 - Regular

Style B - Cube
Style D - Granular

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coating, Exterior, for Tinned Food Cans
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-G-460	-	Glass Containers, Filled and Closed, Packaging and Packing Of

Federal Standards:

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-595	-	Colors

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US Department of Agriculture

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Information Division, Animal and Plant Inspection Service, US Department of Agriculture, Washington, DC 20250.)

US Department of Agriculture

US Standard for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Oil Chemists Society

Official and Tentative Methods

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

American Elsevier Publishing Co., Inc.

Qualitative Analysis by Spot Tests

(Application for copies should be addressed to the American Elsevier Publishing Co., Inc., New York, NY 10017.)

US Army Natick Research and Development Command

Food Technology, Vol. XI, No. 9 (1957) - Hedonic Scale Method of Measuring Food Preferences

(Application for copies should be addressed to the US Army Natick Research and Development Command, ATTN: DRDNA-WTE, Natick, MA 01760.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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National Research Council

Food Chemical Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Ave., N.W., Washington, DC 20418.)

3. REQUIREMENTS

3.1 Bid sample approval. (Unless otherwise specified (see 6.1)) three 1-pound (0.45 kg) samples of the product, which the contractor proposes to furnish, shall be packaged in cans which shall then be hermetically sealed and submitted to the contracting officer prior to the bid opening. One sample of those selected by the contracting officer shall be used to determine bid sample approval by panel testing (see 4.5.5.1). The remaining two samples submitted by the successful bidder shall be used by the inspection service as approved reference samples for examination of deliveries. The approval of any bid samples for palatability will not constitute approval of the samples as meeting the requirements of this specification.

3.2 Materials. The materials shall be clean, sound, wholesome and free from evidence of insect infestation or any other objectionable foreign matter, flavor or odor.

3.2.1 Salt. Salt shall be white, refined sodium chloride, with or without anti-caking agents. Iodized salt and ammonium chloride (NH_4CL) shall not be used.

3.2.2 Sugar. Sugar shall be white, refined granulated cane or beet sugar or a combination thereof.

3.2.3 Brown sugar. Brown sugar shall be derived from cane or beet sugar having a soft grain and covered with a film of cane molasses flavored syrup.

3.2.4 Chicken fat. Chicken fat shall be the product obtained by rendering the wholesome chicken fat of current production under the supervision of the FSQS Poultry and Dairy Div., USDA. The rendered fat may be protected by the types and amounts of antioxidants approved by the FSQS Poultry and Dairy Div., USDA. It shall have a peroxide value of not more than 15 milliequivalents per kilogram of chicken fat.

3.2.5 Dried chicken meat solids. The dried chicken meat solids shall be produced after award of Bouillon (soup and gravy bases) contract in a plant operating under supervision of the FSQS Poultry and Dairy Div., USDA. The dried chicken meat solids can be mechanically deboned meat (MDM) and shall have good flavor and odor, and shall be reduced to a fine powder to provide proper distribution in the finished product.

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3.2.6 Hydrogenated vegetable shortening. The shortening shall be refined hydrogenated vegetable oil of 100 hours or greater stability as determined by the Active Oxygen Method (AOM).

3.2.7 Hydrolyzed vegetable protein (for types I, II and IV). The hydrolyzed vegetable protein shall be obtained by the hydrolysis of any one or any combination of corn, soybean, wheat or yeast proteins from which none of the monosodium glutamate has been extracted during processing, and it shall have a characteristic odor and flavor. The salt-free solids shall be not less than 48.0 percent. The salt content shall be not less than 22.0 nor more than 50.0 percent. Ammonium chloride shall be not more than 1.5 percent. The pH of a 3-percent solution shall be not less than 5.0 nor more than 6.3. (The hydrolyzed protein shall be soluble in water at 90°F (32°C) to the extent of not less than 5 grams in 100 milliliters (ml)).

3.2.8 Monosodium glutamate. Monosodium glutamate shall conform to Food Chemicals Codex.

3.2.9 Onions, dehydrated, powdered. Dehydrated powdered onions shall have the typical odor and flavor of fresh onions when 1 ounce (28 g) of powdered onion is mixed with 2 ounces (57 g) of water (approximately 75°F (24°C)). Alternatively, when the natural onion oil is used, it shall be spray dried, encapsulated with gum arabic and diluted with a dextrose carrier.

3.2.10 Starch. Starch shall be any type of starch that will give the desired results, consistant with the requirements of 3.5.

3.2.11 Dextrin. Dextrin shall be the soluble carbohydrate derived from the partial hydrolysis of starch by dilute acids, enzymes or heat.

3.2.12 Spices and soluble spice flavorings. Spices and spice flavorings shall be true to name and possess characteristic odors and flavors of the type normally used in this product.

3.2.13 Celery flavoring, soluble. Soluble celery flavoring shall be prepared by uniformly dispersing celery seed extractive on a dry, soluble, edible carrier.

3.2.14 Beef extract. The beef extract may be imported or may originate in domestic establishments regularly operated under the supervision of the FSQS Meat Quality Div., USDA. The imported product shall be subject to the current regulations of the FSQS Meat Quality Div., USDA. The beef extract shall contain not more than 18.0 percent moisture, not more than 25.0 percent ash, not more than 5.0 percent salt (as NaCl), not less than 7.0 percent total creatine and creatinine, and not less than 8.0 percent total nitrogen. The beef extract shall show no trace of nitrates or nitrites when tested as specified in 4.5.3.1. The beef extract shall have a characteristic flavor and odor and shall not be scorched or burned.

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3.2.15 Hydrolyzed vegetable protein with added smoke flavor (type III only). The product shall be obtained by the hydrolysis of any one or any combination of corn, soybean, wheat or yeast proteins from which none of the modosodium glutamate has been extracted during processing, and it shall have a characteristic odor and flavor. Natural or artificial smoke flavor shall be added at the levels sufficient to impart a pleasing smoke flavor when used at the formula level designed for the product. The salt free solids shall not be less than 50.0 percent. The pH of a 3-percent solution shall be not less than 4.7 nor more than 6.3. The mixture of hydrolyzed protein and smoke flavor shall be soluble in water at 90°F (32°C) to the extent of not less than 5 grams in 100 ml.

3.2.16 Hydrolyzed and unhydrolyzed vegetable protein (type III only). The product shall be obtained from any one or any combination of corn, soybean, wheat or yeast proteins either hydrolyzed or unhydrolyzed. No portion of the monosodium glutamate may be extracted during processing. It shall have a characteristic odor and a flavor free from objectionable bitterness. The pH of a 3-percent solution shall be not less than 4.9 nor more than 6.3.

3.2.17 Caramel coloring. Caramel coloring shall be a dry, readily-soluble brown powder obtained from sucrose or glucose.

3.2.18 Autolyzed yeast. Autolyzed yeast shall be an extract prepared from whole fresh Saccharomyces cerevisiae or Saccharomyces carlbergensis (either primary dried or debittered brewer's yeast) solely by the action of those proteolytic enzymes naturally present in the fresh yeast. It shall have a typical autolyzed yeast flavor and be free from bitter taste. Sodium chloride may be added in the initial plasmolysis or subsequent processing to facilitate flavor development. Total nitrogen shall be not less than 10.0 percent on a salt-free solids basis. Amino nitrogen shall be not less than 25.0 percent of the total nitrogen. The product shall be soluble in water at 90°F (32°C) to the extent of not less than 5 grams of dry substance in 100 ml. The pH in a 3-percent solution shall be not less than 5.1 nor more than 6.1.

3.2.19 Beef fat. The fat shall be oleo oil derived from pure beef fat produced under the supervision of the FSQS Meat Quality Div., USDA. Antioxidants approved by the FSQS Meat Quality Div., USDA may be added during manufacture. Oleo oil shall have a stability of not less than 100 hours A.O.M.

3.2.20 Disodium inosinate and disodium guanylate. Disodium inosinate and disodium guanylate shall be free flowing powder produced by enzymatic treatment of yeast ribonucleic acid. In addition, disodium inosinate and disodium guanylate shall comply with the additive requirements of the U.S. Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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3.3 Formulas.

3.3.1 Class 1 of types I, II and III soup and gravy bases shall be prepared according to the following formulas:

Military Agencies

Percent by Weight

Material	Type I, Class 1 Beef-Flavored	Type II, Class 1 Chicken-Flavored	Type III, Class 1 Ham-Flavored
Salt		Not more than 35.0 percent	
Chicken Fat		Not less than 5.0 percent	
Dried Chicken Meat Solids		Not less than 1.5 percent	
Hydrolyzed Vegetable Protein	Not less than 18.0 percent		
Beef Extract or Beef Extract Replacement 1/	Not less than 11.5 percent		
Hydrolyzed Vegetable Protein with Added Smoke Flavor			Not less than 15.0 percent
Hydrolyzed and/or Unhydrolyzed Vegetable Protein			Not less than 15.0 percent
Monosodium Glutamate 3/	Not less than 3.3 percent (added)	Not less than 5.0 percent (added)	Not less than 6.0 percent (added)
Disodium Guanylate and Inosinate	Not less than 0.1 percent	Not less than 0.1 percent	Not less than 0.1 percent
Onion Powder or Onion oil	Not less than 6.5 percent Not less than 5.2 percent	Not less than 2.0 percent Not less than 0.8 percent	Not less than 7.0 percent Not less than 2.8 percent
Autolyzed Yeast (salt free solids)	Not less than 3.0 percent		

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Military Agencies

Percent by Weight (cont'd)

Material	Type I, Class 1 Beef-Flavored	Type II, Class 1 Chicken-Flavored	Type III, Class 1 Ham-Flavored
Hydrogenated Vegetable Shortening	Not more than 3.0 percent		Not more than 4.0 percent
Sugar	Not more than 12.0 percent	Not more than 10.0 percent	Not more than 8.0 percent
Brown Sugar			Not more than 12.0 percent
Starches or Dextrines (or a combination thereof)	Not more than 7.5 percent	Not more than 18.0 percent	Not more than 11.0 percent
Soluble Celery Flavoring	Not more than 12.0 percent		
Caramel Coloring	Not more than 1.5 percent		
Salt, Spices or Spice Extractives, Disodium Inosinate, Disodium Guanylate, Imitation Beef Flavoring and Other Seasonings except added Ammonium Chloride	To make up to 100 percent		To make up to 100 percent
Hydrogenated Vegetable Fat, Hydrolyzed Protein, Spices and Flavorings, and Other Seasonings 2/		To make up to 100 percent	

1/ Beef extract replacement shall be a suitable, high quality, hydrolyzed vegetable protein (or combination of hydrolyzed vegetable proteins) which meets the requirements of 3.2.7 and is specifically designed to be used as a substitute for natural beef extract. It shall be used at a level (not less than 11.5 percent) which will result in a finished product having equal or better flavor characteristics than the product made with natural beef extract.

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2/ Turmeric, when used to attain a desirable yellow color shall be of the alleppy type.

3/ Monosodium glutamate available in hydrolyzed vegetable protein during the normal manufacturing process of hydrolyzed vegetable protein shall not be calculated as the required monosodium glutamate specified in the formulation.

Civil Agencies

	Percent by Weight			
Material	Type I, Class 1 Styles A and B	Type I, Class 1 Style C	Type II, Class 1 Styles A and B	Type II, Class 1 Style C
Salt				Not more than 35 percent
Chicken Fat				Not less than 5 percent
Dried Chicken Meat Solids			Not less than 2 percent	
Hydrolyzed Vegetable Protein	Not less than 18.0 percent			
Beef Extract or Beef Extract Replacement 1/	Not less than 11.0 percent			
Monosodium Glutamate	Not less than 6.0 percent			Not less than 10.0 percent
Onion Powder	Not less than 6.5 percent		Not less than 2.0 percent	
Autolyzed Yeast (salt free solids)	Not less than 3.0 percent			
Beef Fat	Not more than 5.0 percent	Not more than 20.0 percent		

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Civil Agencies

Percent by Weight (cont'd)

Material	Type I, Class 1 Styles A and B	Type I, Class 1 Style C	Type II, Class 1 Styles A and B	Type II, Class 1 Style C
Sugar	Not more than 12.0 percent		Not more than 10.0 percent	
Starches or Dextrines (or a combination thereof)			Not more than 10.0 percent	
Soluble Celery Flavoring	Not more than 12.0 percent			
Caramel Coloring	Not more than 1.5 percent			
Salt, Spices or Spice Extractives, or Flavorings or Other Seasonings except added Ammonium Chloride to make up 100 parts. Salt shall not exceed 55 percent	To make up to 100 percent	To make up to 100 percent		
Hydrogenated Vegetable Fat, Hydrolyzed Protein, Spices and Flavorings and Other Seasonings to make up 100 parts.			To make up to 100 percent	To make up to 100 percent

1/ Beef extract replacement shall be a high quality, hydrolyzed vegetable protein (or combination of hydrolyzed vegetable protein) which meets the requirements of 3.2.7 and is specifically designed to be used as a substitute for natural

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beef extract. It shall be used at a level (not less than 11 percent) which will result in a finished product having equal or better flavor characteristics than the product made with natural beef extract.

3.3.2 Soup and gravy base (types I and II, class 2, all styles). The types I and II, class 2, all styles shall be formulated in such a manner that the sodium content shall conform to the requirements of 3.5.2.

3.3.3 Type IV, vegetable protein bouillon (class 1, style B and D). Vegetable protein bouillon product shall be composed of salt, autolyzed yeast, hydrolyzed vegetable protein, monosodium glutamate, hydrogenated vegetable fat, sugar, spices or spice extractives, other seasoning and caramel coloring. The mixture shall consist of approximately 10 percent of autolyzed yeast and not less than 25 percent of hydrolyzed vegetable protein.

3.4 Processing. Drying of components singly or in combination prior to final mixing is permissible.

3.5 Finished product.

3.5.1 Physical requirements.

3.5.1.1 Style A. Style A product shall be a uniform mixture of the ingredients specified in 3.3 for the applicable types and classes of the product. The mixture shall be free of lumps that can not be broken by light finger pressure. Not less than 100 percent shall pass through a U.S. Standard No. 5 sieve.

3.5.1.2 Style B. Style B product shall be in the form of compressed cubes of the ingredients specified in 3.3, for the applicable types and classes of the product. One hundred wrapped cubes shall weigh not less than 14 ounces (397 g) or more than 16 ounces (454 g).

3.5.1.3 Style C. Style C product shall be a uniformly blended product in paste form of the ingredients specified in 3.3 for the applicable types and classes of the product.

3.5.1.4 Style D. Style D product shall be a uniformly blended product in granular form of the ingredients specified in 3.3 for the type IV product. One hundred percent shall pass through a No. 5 U.S. Standard sieve and 100 percent shall be retained on a U.S. Standard No. 10 sieve.

3.5.2 Analytical requirements.

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Percent by Weight

	Civil agencies	Military agencies
Type I, Beef flavor		
Class 1, Styles A and B		
Salt (calculated as NaCl)	Not more than 45.0 percent	Not more than 42.0 percent
Moisture	Not more than 2.0 percent	Not more than 2.25 percent
Sugar (as invert)	Not more than 25.0 percent	Not more than 25.0 percent
Fat	Not more than 10.0 percent	Not more than 5.0 percent
Total nitrogen	Not less than 2.5 percent	Not less than 1.5 percent
Total solids	Not less than 98.0 percent	Not less than 98.0 percent
Type I, Beef flavor		
Class 1, Style C		
Salt	Not more than 45.0 percent	--
Moisture	Not more than 2.0 percent	--
Fat	Not more than 28.0 percent	--
Total nitrogen	Not less than 2.5 percent	--
Total solids	Not less than 98.0 percent	--

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Percent by Weight (cont'd)

	Civil agencies	Military agencies
Type II, Chicken flavor		
Class 1, Styles A and B		
Salt (calculated as NaCl)	Not more than 45.0 percent	Not more than 39.0 percent
Moisture	Not more than 2.0 percent	Not more than 2.5 percent
Fat	Not more than 10.0 percent	Not more than 10.0 percent
Sugar (as invert)	Not more than 30.0 percent	Not more than 30.0 percent
Total nitrogen	Not less than 1.5 percent	Not less than 0.7 percent
Type II, Chicken flavor		
Class 1, Style C		
Salt (calculated as NaCl)	Not more than 45.0 percent	--
Moisture	Not more than 2.0 percent	--
Fat	Not more than 20.0 percent	--
Total nitrogen	Not less than 1.5 percent	--
Type III, Ham flavor		
Class 1, Style A		
Salt (calculated as NaCl)	--	Not more than 32.0 percent

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Percent by Weight (cont'd)

	Civil agencies	Military agencies
Moisture	--	Not more than 2.5 percent
Fat	--	Not more than 5.0 percent
Sugar (as invert)	--	Not more than 32.0 percent
Total nitrogen	--	Not less than 1.5 percent
Type IV, Vegetable protein bouillon		
Class 1, Styles B and D		
Salt (calculated as NaCl)	Not more than 55.0 percent	--
Fat	Not more than 10.0 percent	--
Total nitrogen	Not less than 1.5 percent	--
Total solids	Not less than 96.0 percent	--
Type I, Beef flavor		
Class 2, Styles A and B		
Sodium	Not more than 0.40 percent	--
Moisture	Not more than 3.3 percent	--
Fat	Not more than 9.0 percent	--

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Percent by Weight (cont'd)

	Civil Agencies	Military agencies
Type I, Beef flavor		
Class 2, Style C		
Sodium	Not more than 0.40 percent	--
Moisture	Not more than 3.3 percent	--
Fat	Not more than 25.0 percent	--
Type II, Chicken flavor		
Class 2, Styles A and B		
Sodium	Not more than 0.10 percent	--
Moisture	Not more than 3.3 percent	--
Fat	Not more than 11.0 percent	--
Type II, Chicken flavor		
Class 2, Style C		
Sodium	Not more than 0.10 percent	--
Moisture	Not more than 3.3 percent	--
Fat	Not more than 24.0 percent	--

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3.6 Prepared product. When prepared in accordance with 3.6.1 or 3.6.2, as applicable, the finished product shall readily dissolve when stirred, except that type III prepared product may have a maximum sedimentation of 1/8 inch (3 mm) when tested in accordance with 4.5.4. The prepared product shall have the odor, flavor, and appearance of the pertinent type and class of product and shall be free from rancid, scorched, toasted, sour, caramelized, or other objectionable odor or flavor.

3.6.1 Military agencies. A single serving shall be prepared by adding 1/4 ounce (7 g) of product to 8 ounces (227 g) of freshly boiled water.

3.6.2 Civil agencies. A single serving shall be prepared in accordance with the labeled direction for use.

3.6.3 Palatability (Military). The prepared delivery product shall be equal in palatability to that of the prepared approval bid sample.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.8 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. For Military agencies, the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certificate of compliance.

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4.2 Plant qualification.

4.2.1 For civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 For military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.8.

4.3 Bid sample approval. Bid samples shall be tested along with reference standards as prescribed in 4.5.5.1. A minimum of 24 responses shall be obtained for each bid sample. One or more reference standards shall be randomly assigned to an equal number of test subjects. Any bid sample which has a mean rating significantly lower, at the 5 percent level of significance than that of the reference sample shall not be considered acceptable.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless other excluded, amended, modified, or qualified in this specification, or applicable purchase document.

4.4.1.1 Examination of ingredients. Each ingredient shall be examined organoleptically, as necessary, to determine conformance to 3.2 and as may be contained in the ingredient requirements paragraph. The sample shall be approximately 8 ounces (227 g) of product from each of five primary containers, or each container in a lot if less than five form a lot. Conformance of ingredients to identity requirements (kind, type, class, grade, origin inspection, method of manufacturing or processing, formulation, additives, etc., as applicable) shall be ascertained by examination of pertinent labels, invoices, certificates of analysis or other valid documents. Nonconformance to identity or to condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved end item.

4.4.1.2 Testing of components. Testing of components described in this specification shall be as shown in table I. Results for the characteristics listed shall be applicable to the lot average. Nonconformance to one or more test requirements shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

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TABLE I. Testing of components (see 4.4.1)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para 4/
Chicken fat	One pound (0.45 kg) composite 1/	Pounds	S-1	Peroxide value	Nearest milli-equivalent	3.2.4 4.5.3
Beef fat (Oleo oil)	One pound (0.45 kg) composite 1/	Pounds	S-1	Stability (AOM)	Pass or fail 2/	3.2.19 4.5.3
Hydrogenated Veg. shortening	One pound (0.45 kg) composite 1/	Pounds	S-1	Stability (AOM)	Pass or fail 2/	3.2.6 4.5.3
Hydrolyzed veg. protein	One-half pound (0.23 kg) composite 1/	Pounds	S-1	pH	Nearest 0.1	3.2.7 4.5.3
				Salt content	Nearest 0.1 percent	3.2.7 4.5.3
				Salt free solids	Nearest 0.1 percent	3.2.7 4.5.3
				Solubility	Pass or fail	3.2.7 4.5.3
				Ammonium chloride	Nearest 0.1 percent	3.2.7 4.5.3

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TABLE I. Testing of components (see 4.4.1) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para 4/
Beef extract	One-half pound (0.23 kg) composite 1/	Pounds	S-1	Moisture	Nearest 0.1 percent	3.2.14 4.5.3
				Ash	Nearest 0.1 percent	3.2.14 4.5.3
				NaCl	Nearest 0.1 percent	3.2.14 4.5.3
				Creatine and Creatinine	Nearest 0.1 percent	3.2.14 4.5.3
				Nitrogen	Nearest 0.1 percent	3.2.14 4.5.3
				Nitrites and Nitrates	Pass or fail	3.2.14 4.5.3.1
Hydrolyzed veg. protein w/ smoke flavor	One-half pound (0.23 kg) composite 1/	Pounds	S-1	pH	Nearest 0.1 percent	3.2.15 4.5.3
				Solubility	Pass or fail	3.2.15
				Salt free solids	Nearest 0.1 percent	3.2.15 4.5.3

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TABLE I. Testing of components (see 4.4.1) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Character-istics	Results reported	Test ref. and rqmt. para 4/
Hydrolyzed and un-hydrolyzed veg. pro-tein	One-half pound (0.23 kg) composite 1/	Pounds	S-1	pH	Nearest 0.1 percent	3.2.16 4.5.3
Autolyzed yeast	Four ounce (113 g) composite 1/	Pounds	S-1	Nitrogen (salt free solids basis)	Nearest 0.1 percent	3.2.18 4.5.3
				Amino nitrogen	Nearest 0.1 percent	3.2.18 4.5.3
				pH	Nearest 0.1 percent	3.2.18 4.5.3
				Solu-bility	Pass or fail	
Cans 3/	One can and lid	Cans	S-1	Tinplate	Nearest 0.01 lb. per base box	5.1.1.1 4.5.6

1/ The composite shall be derived from the number of primary containers indicated by the inspection level.

2/ If failure, report to nearest hour.

3/ Testing of exterior can coating (when required) and can label shall be performed on the same cans submitted for tinplate test.

4/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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4.4.2 Certification.

4.4.2.1 USDA Certificate of Quality and Condition. When specified by the procuring agency, the supplier shall furnish a U.S. Department of Agriculture Certificate of Quality and Condition as outlined in procurement documents (see 6.1).

4.4.3 In-process inspection.

4.4.3.1 Wholesomeness. When specified by the procuring agency, the bouillon shall be prepared under the continuous inspection service of the applicable agency of the U.S. Department of Agriculture. The immediate containers and shipping containers of beef extract and chicken-flavored bouillon shall bear the official inspection mark or legend of that inspection agency (see 6.1).

4.4.3.2 Examination of formulation. Unless otherwise specified (see 6.1) examination shall be performed to determine conformance to formulation requirements of 3.3, as applicable. Records of formulation shall be maintained. In the event the Government inspector is in the plant when production is in progress, he shall perform an examination of the actual formulation. Nonconformance shall be cause for rejection of the involved finished product.

4.4.4 End item inspection.

4.4.4.1 End item examination. Examination shall be in accordance with tables III through VIII. Sampling plans for the above referenced tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables V, VI and VIII and percent defective for tables III, IV and VII.

TABLE II. Sampling plans

Table	Inspection level	Sample unit	Expression of lot size	AQLs Major A	Major B	Minor
III	S-3	One filled primary container	Primary containers	-	-	2.5
IV	S-3	One filled and closed envelope	Envelopes	-	1.0	-
V	S-2	One filled envelope or 4 oz. (113 g) from can or jar	Primary containers	1/	1.5	-

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TABLE II. Sampling plans (cont'd)

Table	Inspection level	Sample unit	Expression of lot size	AQLs Major A	Major B	Minor
VI	S-2	One filled and closed intermediate paperboard carton	Cartons	-	-	2.5
VII	S-2	One filled carton, bulk packed shipping container or can	Cartons, shipping containers or cans	-	-	2.5
VIII	I	One filled and closed jar or envelope	Jars or envelopes	.25	1.5	6.5

1/ The finding of any major A defect shall be cause for rejection of the lot.

TABLE III. Examination for net weight

Category	Defect
Minor	
201	More than 5 percent under specified net weight (for containers required to have 8 oz. (227 g) or less). 1/
202	More than 2 percent under specified net weight (for containers required to have 1 pound (.45 kg) or more). 1/
203	Net weight of cubes less than 14 ounces (397 g) or more than 16 ounces (454 g).

1/ Report results to nearest 1/16 oz. (nearest gram) for "8 oz." containers; nearest 0.1 gram for "7 gram" envelopes; and nearest 1/8 oz. (nearest gram) for container required to have one pound (454 g) or more. (Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.)

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TABLE IV. Examination for envelope delamination and leakage 1/

Category	Defect
Major B	
151	Delamination
152	Leakage

1/ Examine for leakage and delamination i.e. separation of plys after removal from water (see 4.5.2).

TABLE V. Examination of product characteristics

Category		Defect
Major A	Major B	Finished product
	151	Not uniformly mixed (powdered, granular or paste).
	152	Not free from lumps that do not break up with light finger pressure (powdered product).
	153	Cubes not completely wrapped.
		Prepared product 1/
	154	Does not dissolve readily when stirred (except type III product).
	155	Palatability not equal to bid sample.
	156	Odor or appearance not characteristic of the product.
101		Rancid, sour, scorched, toasted, caramelized or other objectionable flavor or odor, shall be cause for rejection of the lot.
102		Presence of any foreign material (e.g., insects, filth, glass, wood, metal, paint, etc.) shall be cause for rejection of the lot.

1/ Make one individual serving per sample unit in accordance with 3.6.1 or 3.6.2, as applicable.

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TABLE VI. Examination of intermediate paperboard box

Category	Defect
Minor	
201	Box not as specified. 1/
202	Box dirty.
203	Box not securely closed with tape. 1/
204	Arrangement of envelopes within box not as specified.
205	Box divider missing.
206	Marking on intermediate box missing, incorrect or illegible.

1/ Measuring device shall be calibrated in 1/16 inch (2 mm) increments.

TABLE VII. Count of envelopes and cubes

Category	Defect
Minor	
201	Less than 100 cubes per can.
202	Less than 200 envelopes per intermediate box (level A).
203	Less than 1000 envelopes per bulk packed shipping container or box (level C).

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TABLE VIII. Examination of jars and envelopes (Military)

Category			Defect
Major A	Major B	Minor	
			Jars
101			Broken or cracked.
	151		Chipped or thin spot.
	152		Pitted rust on cap. 1/
	153		Cap loose. 2/
		201	Cap cross threaded.
		202	Not clean.
	154		Cap liner missing or does not form a tight seal against mouth of jar.
			Envelopes
	155		Style or size not as specified. 3/
102			Tear, hole, open seal, or sifter.
	156		Color not as specified.
	157		Delamination.
	158		Not clean.
			Jars and envelopes
	159		Nomenclature missing, incorrect or illegible.
	160		Directions for use (when required) missing, incorrect or illegible.
		203	Other required labeling information missing, incorrect or illegible.

1/ Rust that can be removed by wiping with a soft cloth shall not be considered pitted, rust.

2/ For determining tightness of screw type cap, the following is applicable:

Place the jar upright on top of filled and closed shipping container. Grasp the sides of the cap firmly with fingertips of one hand and turn cap gently counter-clockwise with a minimum of downward pressure for one complete revolution. If cap turns and jar remains stationary, a defect shall be scored.

3/ Measurement device shall be calibrated in 1/16 inch (2 mm) increments.

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4.4.4.2 Examination of particle size for finished product (style A for Military - style D for Civil agencies). The finished product shall be examined for particle size in accordance with sieving procedures detailed in 4.5.1. The sample shall be a 100 gram composite derived from the number of primary containers (except envelopes) indicated by inspection level S-2. Lot size shall be expressed in terms of the primary containers. A sufficient quantity of 7 gram envelopes shall be utilized to form a 100 gram composite. Less than 100 percent (to nearest whole percent) passing through the specified sieve shall be cause for rejection of the lot.

4.4.4.3 End item testing. The finished product shall be tested to determine conformance to the analytical requirements listed in the tabulated data of 3.5.2. In addition, type III product shall be tested to determine conformance to sedimentation requirement of 3.6. Test procedures shall be in accordance with 4.5. The sample for testing (except envelopes) shall be an eight-ounce composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of the primary containers. The sample for product in envelopes shall be a composite derived from 32 envelopes. Each result shall be reported to the same decimal point as specified for the pertinent requirement except for sedimentation which shall be reported as pass or fail. A test failure shall be cause for rejection of the lot.

4.4.5 Packaging inspection.

4.4.5.1 External examination of cans. External examination of cans shall be performed in accordance with the applicable examination criteria of PPP-C-29.

4.4.5.2 Examination of can coating. When cans are required to be coated in accordance with TT-C-495, examination of coating shall be performed in accordance with the applicable quality assurance provisions of TT-C-495.

4.4.5.3 Examination for can labeling. Examination for labeling shall be performed in accordance with examination criteria of MIL-L-1497.

4.4.5.4 Examination of shipping containers packed with glass jars. Examination shall be performed in accordance with the applicable quality assurance provisions of PPP-G-460.

4.4.5.5 Examination of shipping containers packed with cans or flexible packages. Examination of the filled, closed and reinforced shipping containers shall be in accordance with the applicable examination criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: National Stock Number, item description, contract number of date of pack markings missing, incorrect or illegible. Minor: (1) Other required markings missing, incorrect or illegible; (2) Number and arrangement of cans not as specified; (3) Number and arrangement of intermediate boxes (containing flexible packages) not as specified; (4) When applicable, reinforcement with non-metallic strapping or tape is not used. Level C shipping containers shall be examined only for the aforementioned marking defects and for the closure method specified in 5.2.3.

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4.4.5.6 Examination of unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.4.5.7 Flexible packages. Conformance of envelope material to thickness requirements of 5.1.1.2 shall be ascertained by examination of pertinent invoices, labels, or other valid documents from the supplier of envelope material. Non-conformance shall indicate unacceptable material and use of such shall be cause for rejection of the involved end item.

4.5 Methods of inspection.

4.5.1 Sieving. Sieving shall be performed as follows: Place a 100-gram sample on top of specified sieve or specified assembled sieves, as applicable, with attached pan. Attach cover and with left hand hold sieve assembly at an angle (approximately 30 degrees from the horizontal) so that upper edge of the tilted plane is toward the right hand. Shake by striking the right side of the assembly with a short sharp stroke with right hand, moving in a horizontal plane. After each 25 strokes, turn sieve assembly one-sixth of a revolution in same direction. Continue shaking for 5 minutes.

4.5.2 Leakage. The filled and sealed flexible package shall be examined for leakage by submerging in water, contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (67 kPa) (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leakage.

4.5.3 Chemical analyses. Chemical analyses, when required, shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists, except as otherwise indicated:

Test	Source	Method
Beef extract:		
Moisture	Chapter: Fruits and Fruit Products	Moisture in Dried Fruits 1/
Ash	Chapter: Meat and Meat Products Section: Meat Extracts and Similar Products	Ash

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Test	Source	Method
Beef extract: (cont'd)		
Salt (NaCl)	Chapter: Meat and Meat Products, Section: Meat Extracts and Similar Products	Chloride
Creatine	Chapter: Meat and Meat Products, Section: Meat Extracts and Similar Products	Creatine
Creatinine	Chapter: Meat and Meat Products, Section: Meat Extracts and Similar Products	Creatinine
Total nitrogen	Chapter: Meat and Meat Products, Section: Meat Extracts and Similar Products	Total nitrogen
Hydrolyzed and unhydrolyzed vegetable protein:		
Ammonium chloride	Chapter: Fertilizer Section: Nitrogen	Ammonical (Magnesium Oxide Method)
Moisture	See test under beef extract	
Salt	See test under beef extract	
pH	Chapter: Beverages: Malt Beverages, and Brewing Materials, Section: Beer	Hydrogen-ion Concentration (pH)
Chicken fat		
Peroxide value	2/	Cd-8-53

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Test	Source	Method
Hydrogenated shortening, and beef fat		
Stability (active oxygen method)	2/	Cd-12-57
Autolyzed yeast		
Total nitrogen	See test under beef extract	
Amino nitrogen	Chapter: Meat and Meat Products, Section: Meat	Sorensen Method
pH	See test under hydrolyzed and unhydrolyzed vegetable protein	
Finished product		
Total chlorides	See test under beef extract	
Moisture	See test under beef extract	
Total fat	Chapter: Meat and Meat Products, Section:	Crude Fat or Ether Extract
Total sugars 3/ (as invert)	Chapter: Sugar and Sugar Products, Section: Sugar and Sirup	Sucrose-Chemical Methods (reducing sugar invert)
Sodium content	Chapter: Fish & Other Marine Products	Sodium and Potassium-Official Final Action
Total nitrogen	See test under beef extract	

1/ Drying time shall be six hours at 70°C (158°F), and the pressure shall not exceed 50 mm of mercury.

2/ Official and Tentative Methods of the American Oil Chemists' Society.

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3/ Remove fat from 10 grams of soup and gravy base as in AOAC, Chapter: Beverages: Malt Beverages and Brewing Materials, Cereal Adjunct (Oil or Petroleum Extract). The resultant powder is diluted as follows: Dilution-10 gm/250 ml. H_2O, 50 ml/200 ml, H_2O, 50 ml aliquot, proceed with analysis: Sucrose Chemical Method (reducing sugar invert).

4.5.3.1 Nitrates and nitrites. Qualitative tests for nitrates and nitrites shall be made as specified in Spot Tests in Inorganic Analysis by Feigl: Chapter 4. Tests for Acid Radical; method 25-Nitric Acid and method 26-Nitrous Acid.

4.5.4 Sedimentation. Sedimentation shall be determined as follows; to a 400 mL beaker, containing 240 mL of hot water 180°F (82°C), add gradually 7 grams of ham-flavored soup and gravy mix, and mix well. Measure sedimentation after the mixture has been standing 5 minutes after reaching a temperature of 70°F (21°C).

4.5.5 Palatability. Tests for palatability, as required, shall be performed in accordance with 4.5.5.1 and 4.5.5.2, as applicable.

4.5.5.1 Procedure. Samples shall be prepared as described in 3.6. Approximately 2-ounce (59 mm) portions shall be served, at a temperature of 140° to 145°F (60° to 63°C), in coded containers, to a consumer-type panel. The panel members shall rate each sample on a 9-point hedonic scale* graduated in successive degrees of like and dislike. The sample shall be presented successively with a minimum interval of 30 seconds between samples, in controlled orders, so that each sample is served an equal number of times in each temporal position. Each subject shall rate from two to six samples at a test session. The subjects shall test without interference from each other or from outsiders.

*NOTE: For further details, see Hedonic Scale Method of Measuring Food Preferences, reprinted from Food Technology, 1957, Vol. XI, No. 9, Supplement Pages 9-14.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

4.5.5.2 Selection of reference standards. Samples complying with requirements of this specification will be tested at the same sessions by a consumer panel yielding not less than 36 ratings per sample. Tests will be conducted according to 4.5.5.1. Samples which are acceptable and can be used for reference standards shall be those samples that are not significantly different from the highest ranking

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at the 5 percent level of significance. Reference samples shall be stored in closed containers and held at a temperature of approximately 40°F (4°C). The reference standards for any given test session shall be selected randomly from among the available reference standards.

4.5.6 Tin coating weight. Tin coating weight shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A. The product shall be preserved in accordance with 5.1.1.1 or 5.1.1.2.

5.1.1.1 Style A product packaged in cans. The style A product shall be filled into a size 401 by 411 can. The net weight shall be 1 pound 8 ounces (680 g). A minus 2 percent tolerance will be allowed in any one container, provided the average net weight of the can inspected in accordance with table III is not less than 1 pound 8 ounces (680 g). The cans shall be coated overall on the inside with an enamel suitable for the product furnished by the can manufacturer. The can shall be an open top style, round, metal can with welded or soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25 pound per base box electrolytic tinplate (see 4.5.6) and shall be coated overall on the outside with a coating conforming to type I of TT-C-495. The can shall be hermetically sealed. The seams shall not leak when subjected to the seam leakage procedure of PPP-C-29.

5.1.1.2 Style A product packaged in flexible packages. The product shall be filled into an envelope which shall be heat sealed along all four sides. The net weight shall be 7 grams. A minus 5 percent tolerance will be allowed in any one envelope, provided the average net weight of the envelope inspected in accordance with table III, is not less than 7 grams. The dimensions of heat-sealed envelope containing the product, shall be not more than 2-5/8 by 2-5/8 inches (67 by 67 mm), or 2-1/2 by 3-1/2 inches (64 by 89 mm). The filled and sealed envelope shall be made from a heat-sealable, laminated material, one lamina of which shall be 0.00035 inch (0.009 mm) thick aluminum foil. The envelope shall show no leakage when examined as specified in 4.5.2. The combined sheet shall show no evidence of delamination (see table IV) when made into envelopes or heat-sealed. The outside of the complete flexible package shall be a dull brown or light yellow.

5.1.1.2.1 Intermediate box for flexible packages. Two hundred envelopes shall be placed on edge in two rows of 100 each in a snug fitting folding paperboard box made in accordance with variety 1, style III, type G, class 1 of PPP-B-566. Paperboard for the box shall be 0.022-inch (0.6 mm) thick, and shall have a minimum bursting strength of 66 pounds (30 kg) per square inch (25 mm^2). A divider shall be used to separate the two rows. Each box shall be securely closed with tape as specified in the appendix of PPP-B-566.

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5.1.2 Level C. The product shall be preserved in accordance with 5.1.2.1, 5.1.2.2, or 5.1.2.3.

5.1.2.1 Style A product packaged in cans. One pound 8 ounces (680 g) of style A product shall be unit packed in accordance with 5.1.1.1, except that cans with or without commercial exterior coating will be acceptable.

5.1.2.2 Types I and II, styles A, C and type IV, style D product packaged in glass jars or metal cans. The product shall be filled into a glass jar or metal can. The net weight of glass jars as specified shall be either 8 ounces (227 g) or 1 pound (0.45 kg). A minus 5 percent for 8 ounces (227 g) or minus 2 percent for 1 pound (0.45 kg) tolerance will be allowed in any one jar, provided the average net weight shall conform to requirements of table III. The glass jar shall be that normally used by the supplier for the product and shall be furnished with a metal or plastic screw cap closure and a tight fitting liner. The net weight of cans shall be one pound (0.45 kg). A minus 2 percent tolerance will be allowed in any one can, provided the average net weight of the can inspected in table III is not less than one pound (0.45 kg). The cans shall be in accordance with type III or V, class 1 of PPP-C-96.

5.1.2.3 Types I, II, and IV, style B product packaged in cans. Each cube shall be wrapped in a sheet of material normally used for the product. One hundred cubes shall be packaged in a metal can as specified in 5.1.2.2.

5.2 Packing. Twelve one pound (.45 kg) cans of style A, C or D product or 12 cans of 100 cubes, style B product, arranged 4 in length, 3 in width and 1 in depth or twenty-four 1-1/2 pound (.68 kg) cans of style A product, arranged 4 in length, 3 in width, and 2 in depth, or 8 intermediate boxes of flexible packages, arranged in 2 rows of 4 each, shall be packed in a snug fitting shipping container in accordance with level A, B, or C, as specified (see 6.1). Glass jars of the product shall be packed in accordance with PPP-G-460 for the level specified (see 6.1). When specified (see 6.1), 1000 flexible packages shall be packed directly into a shipping container in accordance with 5.2.3.

5.2.1 Level A. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V2s, of PPP-B-636.

5.2.2 Level B. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V3c, V3s, or V4s, of PPP-B-636.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3) the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape in accordance with the appendix of PPP-B-636, except that one lengthwise and one girthwise band shall be applied.

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5.2.3 Level C. The product shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable except fiberboard containers shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078, for the type and class of load specified (see 6.1).

5.4 Labeling and marking.

5.4.1 Military agencies.

5.4.1.1 Cans. The cans shall be labeled in accordance with MIL-L-1497, with the following information, and as specified in 5.4.1.1.1;

SOUP AND GRAVY BASE, BEEF OR CHICKEN OR IMITATION HAM-FLAVORED, (as applicable)

(in letters larger than any other others used for the label)

Ingredient statement (to appear directly beneath and contiguous to the name of the product)

Net weight_____
Est. No._____(may be embossed on one end of the can, if applicable)
Inspection legend.
Name and address of manufacturer.

5.4.1.1.1 Directions for use. The following directions for use, as applicable, shall be included on the label.

I. For the 8-ounce jar

Stir contents into 2 gallons boiling water. Make 2 gallons stock or 32 one-cup servings.

For one quart, stir 1 ounce (3 tablespoons) base into 1 quart boiling water.

For one cup, stir 2 teaspoons base into 1 cup boiling water.

Use for stock in recipes for soups, gravies, and sauces. Check seasoning before adding salt.

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II. For 1 pound 8-ounce can

Stir contents into 6 gallons boiling water. Makes 6 gallons stock or 96-one cup servings.

For one gallon stir 4 ounces (12 tablespoons) base in 1 gallon boiling water.

For 1 cup, stir 2 teaspoons base into 1 cup boiling water.

Use for stock in recipes for soups, gravies, and sauces.

Check seasoning before adding salt.

III. For 1 pound jar

Stir contents into 4 gallons boiling water. Makes 4 gallons stock or 64 one-cup servings.

For one quart, stir 1 ounce (3 tablespoons) base into 1 quart boiling water.

For one cup, stir 2 teaspoons base into 1 cup boiling water.

Use for stock in recipes for soups, gravies, and sauces.

Check seasoning before adding salt.

IV. Bouillon cubes

For soup - dissolve one cube in 1/3 canteen cup (8 fluid ounces) of hot water.

5.4.1.2 Flexible packages. The following information complying with 5.4.1.1, shall be printed on the envelope. In addition the labeling information shall comply with applicable requirement specified in 5.4.1.1. Printing shall be in a contrasting color to allow easy and legible reading.

SOUP AND GRAVY BASE, BEEF-FLAVORED, CHICKEN OR IMITATION HAM-FLAVORED, (as applicable)

(in letters larger than any others used for the label)

Ingredient statement.

Net weight - 7 grams.

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For soup - dissolve contents in 1/3 canteen cup (8 fluid ounces) of hot water.

Name and address of manufacturer.

5.4.1.3 Glass jars. Commercial labeling that complies with the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder will be acceptable.

5.4.1.4 Intermediate boxes and shipping containers. In addition to any special marking required by the contract or order, intermediate boxes and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1.5 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

5.4.2 Civil agencies.

5.4.2.1 Cans. The cans shall be labeled in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder, and shall include the following information:

Name of item (as applicable to the type Natural or Imitation flavored); style (powder, cube, paste or granular - as applicable); net weight; name and address of manufacturer; directions for use (individual serving and multiple servings) and nutritional information (Class 2 - low sodium product only). In addition, the label for Class 1 - regular product shall include the following: "Note: This product has a high salt content, if used as a soup or gravy base, reduce seasoning accordingly."

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and style of product required (see 1.2).
(c) When other than three 1-pound (0.45 kg) samples of bid sample is required (see 3.1).
(d) When certificate of compliance is not authorized (see 4.4.1).
(e) When pre-acceptance inspection is required by Civil agencies (see 4.2.1).
(f) When USDA certificate of quality and condition is required (see 4.4.2.1).

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(g) Official inspection mark on the container, when required (see 4.4.3.1).
(h) When plant inspection for civil agencies is required (see 4.2.1).
(i) When examination of formulation is not required by civil agencies (see 4.4.3.2).
(j) Applicable levels of preservation and packing required (see 5.1 and 5.2).
(k) Level of packing of glass jars in accordance with PPP-G-460 (see 5.2).
(l) When 1000 flexible packages are to packed directly into shipping container (see 5.2).
(m) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(n) Type and class of unit load, when unit loading is specified (see 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipping, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS
AGR-CMS
HEW-NIH, FDA

Project No. 8935-0061

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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